제 5 장 교류전압 제어기

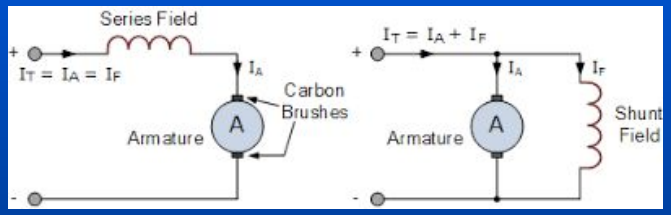
AC to AC Converter

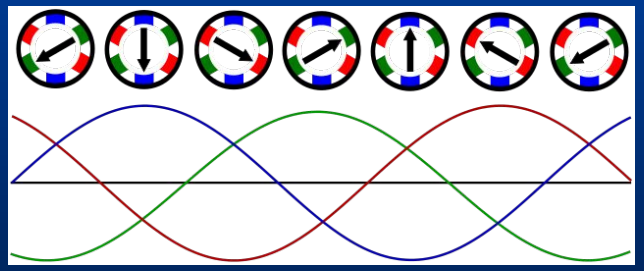
AC→ AC

5.1 개요

- 교류전력을 직접 변환
- Light Dimmer, Small Power Induction Motor 등에 사용
- 종류
  - Phase Control 방식
  - Integral Control 방식

전동기의 종류

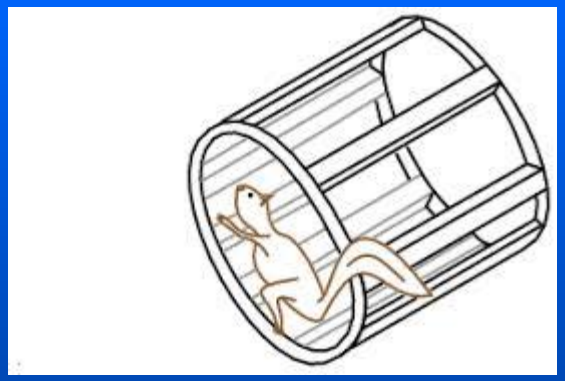


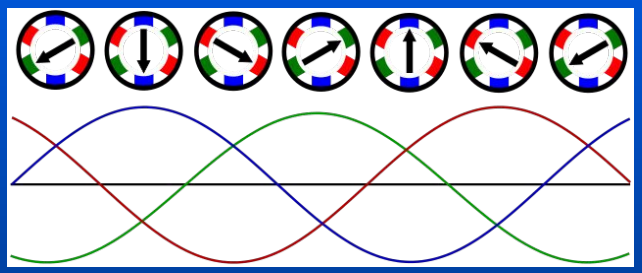


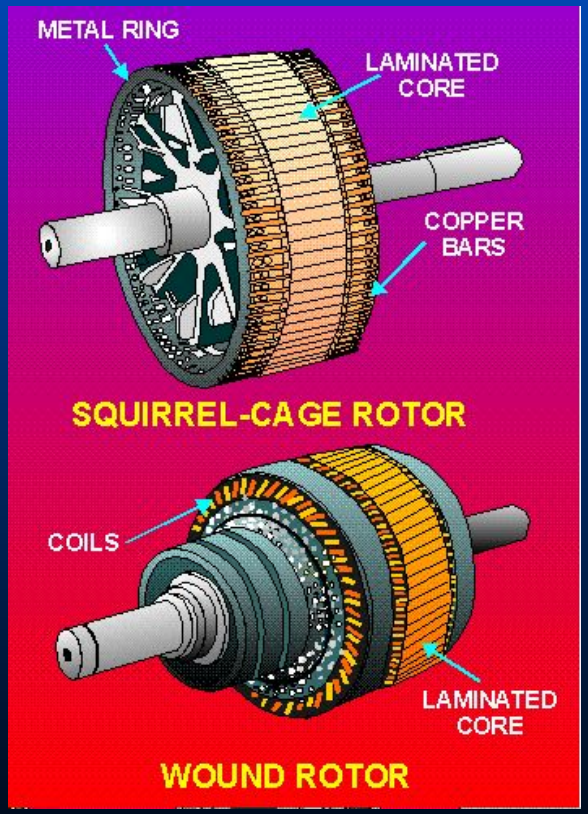
$$N = \frac{120f}{P}(1-s)[rpm]$$

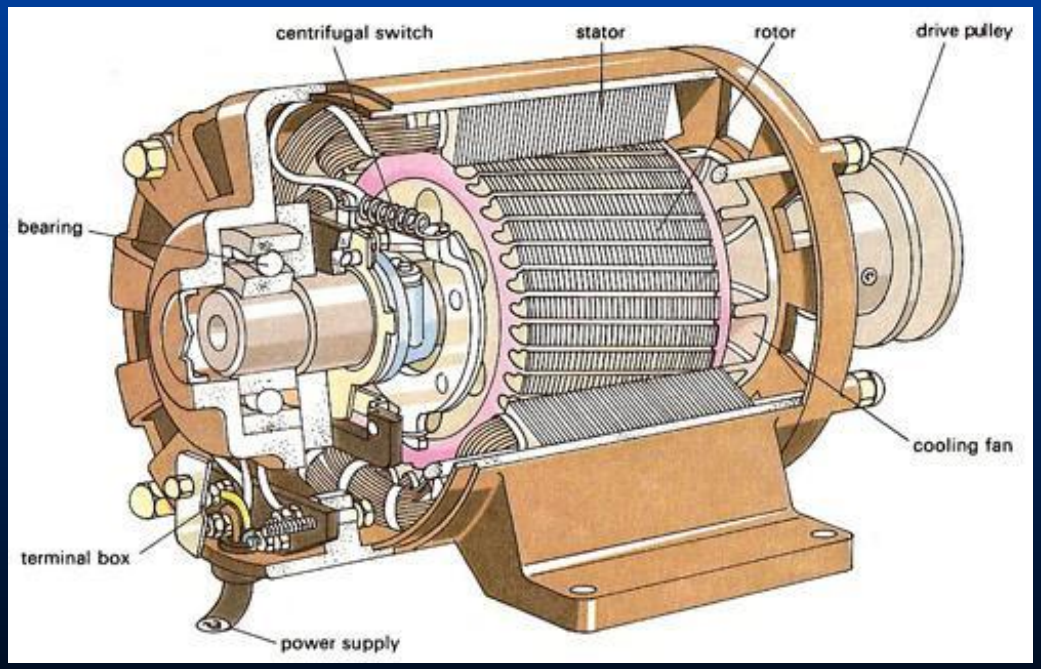
- 직류
  - -분권(shunt): 회전자와 계자 병렬, 정속도
  - -직권(series): 회전자와 계자 직렬, 회전력 (torque)
  - -복권(compound)
- 교류 단상: 회전자계가 발생하지 않음
  - -분상기동, 콘덴서 기동, 반발기동
- 교류 3상: 회전자계가 발생
  - -유도전동기(induction): slip이 발생
    - 암페어의 법칙: 회전자에 자계 유도
    - 농형(squirrel cage): 5kw 이하
    - 권선형(wound) : 5kw 이상
  - -동기전동기(synchronous): 동기속도

3상 유도전동기

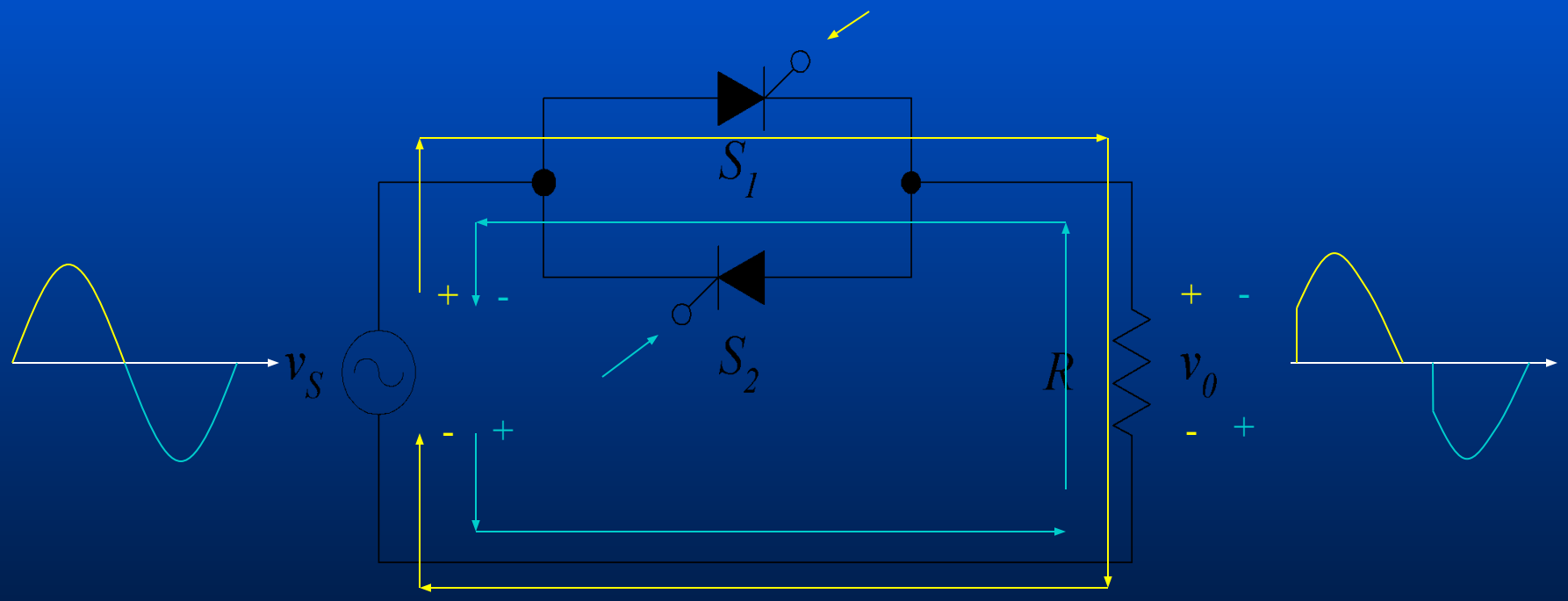




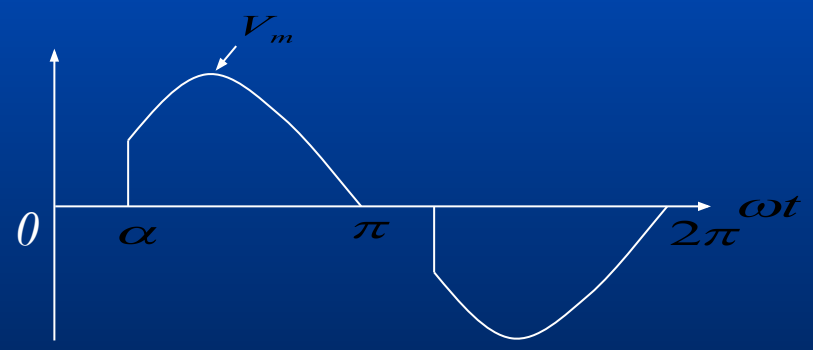

METAL RING
LAMINATED CORE
COPPER BARS
SQUIRREL-CAGE ROTOR
COILS
LAMINATED CORE
WOUND ROTOR


centrifugal switch
stator
rotor
drive pulley
bearing
cooling fan
terminal box
power supply

5.2 단상 교류 전압제어기



5.2 단상교류전압제어기: 실효값



$$V_{o,rms} = \sqrt{\frac{1}{\pi}\int_{\alpha}^{\pi}(V_m \sin\omega t)^2\, d(\omega t)}$$

$$\left(\sin^2\omega t = \frac{1-\cos 2\omega t}{2}\right)$$

$$= \sqrt{\frac{V_m^{\ 2}}{\pi}\left[\frac{\omega t}{2} - \frac{\sin 2\omega t}{4}\right]_{\alpha}^{\pi}}$$

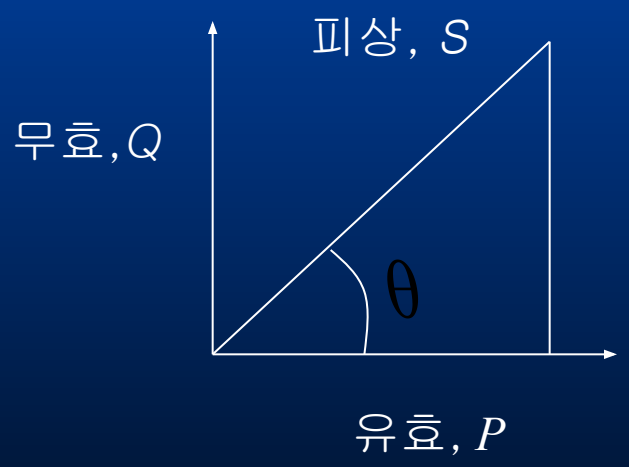
$$= \sqrt{\frac{V_m^{\ 2}}{\pi}\left(\frac{\pi}{2} - \cancel{\frac{\sin 2\pi}{4}} - \frac{\alpha}{2} + \frac{\sin 2\alpha}{4}\right)}$$

$$= \frac{V_m}{\sqrt{2}}\sqrt{\left(1 - \frac{\alpha}{\pi} + \frac{\sin 2\alpha}{2\pi}\right)}$$

5.2 단상교류전압제어기: 역률

- 역률(Power Factor)
  - 역률=유효/피상

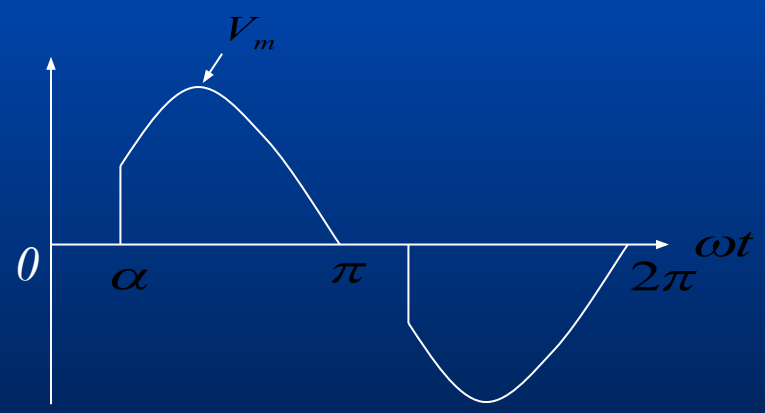
$$pf = \frac{P}{S} = \frac{V_{o,rms}}{V_{s,rms}}$$



$$= \frac{\frac{V_m}{\sqrt{2}}\sqrt{\left(1-\frac{\alpha}{\pi}+\frac{\sin 2\alpha}{2\pi}\right)}}{\frac{V_m}{\sqrt{2}}}$$

$$= \sqrt{\left(1-\frac{\alpha}{\pi}+\frac{\sin 2\alpha}{2\pi}\right)}$$

5.2 단상교류전압제어기: 전류



- 평균값

$$I_{SCR,avg} = \frac{1}{2\pi}\int_{\alpha}^{\pi}\frac{V_m}{R}\sin\omega t\, d(\omega t)$$

$$\int \sin dt = -\cos t$$

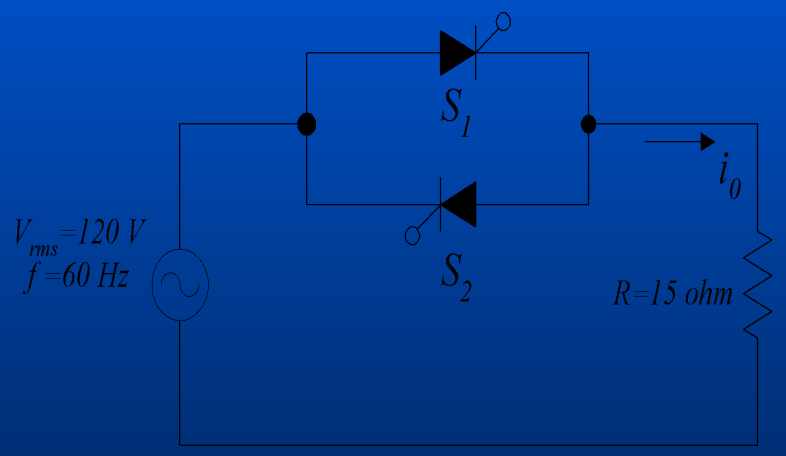
$$= \frac{V_m}{2\pi R}\left[-\cos\omega t\right]_{\alpha}^{\pi}$$

$$= \frac{V_m}{2\pi R}\left(1+\cos\alpha\right)$$

- 실효값

$$I_{base} = \frac{V_{S,rms}}{R}$$

예제 5.1 저항부하를 가진 단상제어기(1)



(a) 500 W를 공급하는 지연각

$$P = \frac{V_{0,rms}^2}{R}$$

$$V_{0,rms} = \sqrt{PR} = \sqrt{(500 \times 15)} = 86.6V$$

$$V_{0,rms} = \frac{V_m}{\sqrt{2}}\sqrt{\left(1 - \frac{\alpha}{\pi} + \frac{\sin 2\alpha}{2\pi}\right)}$$

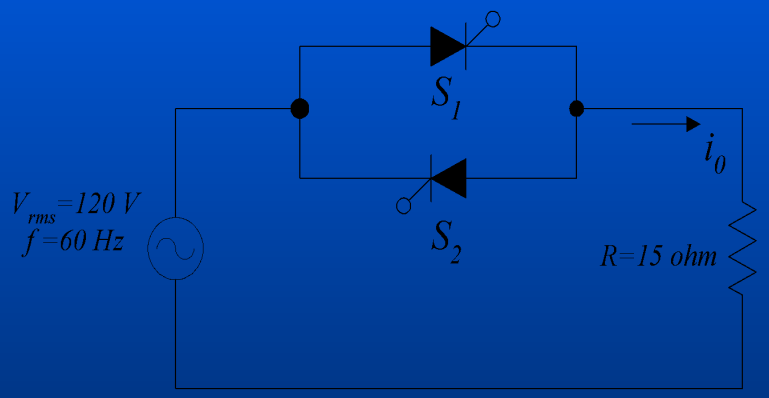
$$\therefore 86.8 - 120\sqrt{\left(1 - \frac{\alpha}{\pi} + \frac{\sin 2\alpha}{2\pi}\right)} = 0$$

$$\alpha = 1.54\,rad = 88.1^o$$

(b) 전원전류의 실효값

$$I_{0,rms} = \frac{V_{0,rms}}{R} = \frac{86.6}{15} = 5.77\,A$$

예제 5.1 저항부하를 가진 단상제어기(2)



(c) SCR 전류의 실효, 평균값

$$I_{SCR,rms} = \frac{I_{0,rms}}{\sqrt{2}} = \frac{5.77}{\sqrt{2}} = 4.08\,A$$

$$I_{SCR,avg} = \frac{V_m}{2\pi R}(1+\cos\alpha)$$

$$= \frac{\sqrt{2}(120)}{2\pi(15)}(1+\cos(88.1)) = 1.86\,A$$

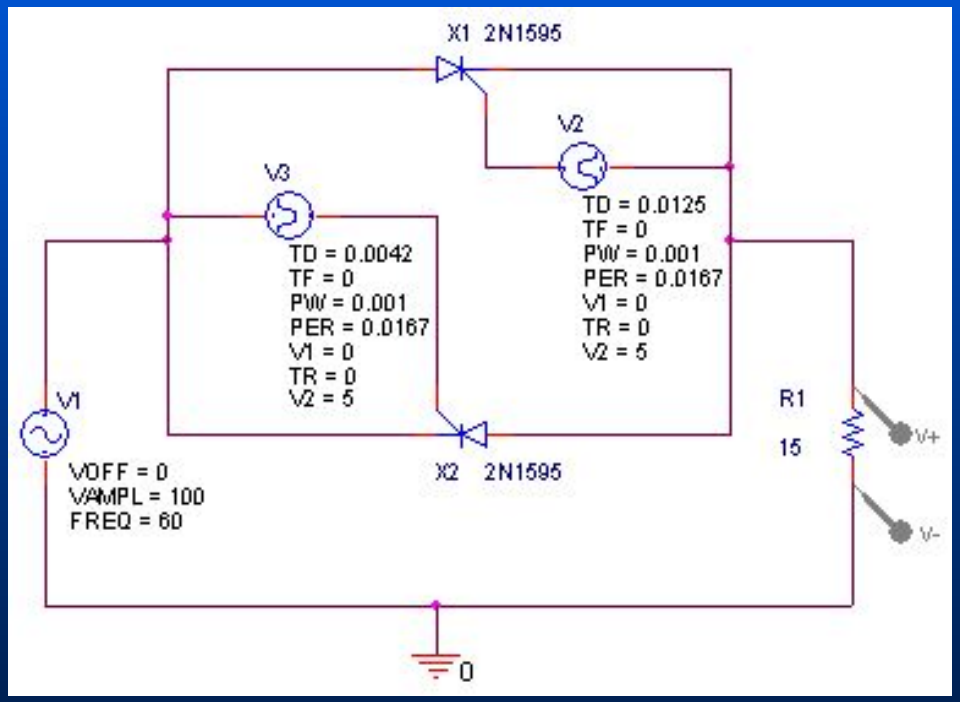
(d) 역률

$$pf = \frac{P}{S} = \frac{500}{(120)(5.77)} = 0.72$$

(e) 전원전류의 실효값

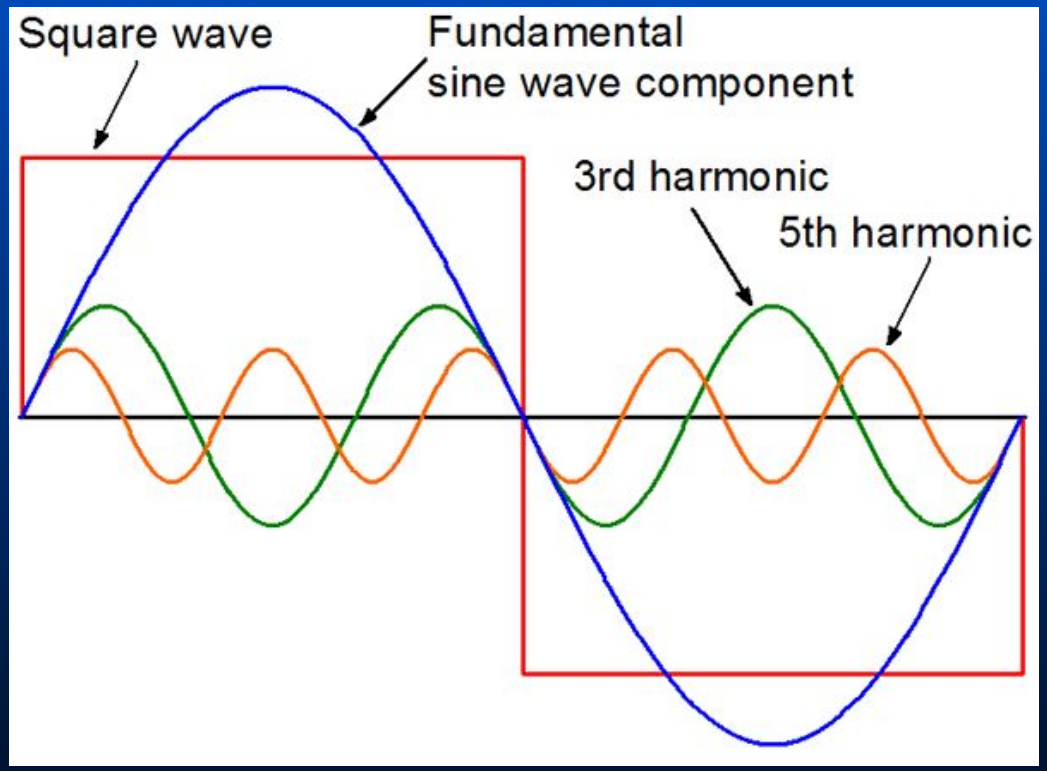
$$I_{base} = \frac{V_{S,rms}}{R} = \frac{120}{15} = 8.0\,A$$

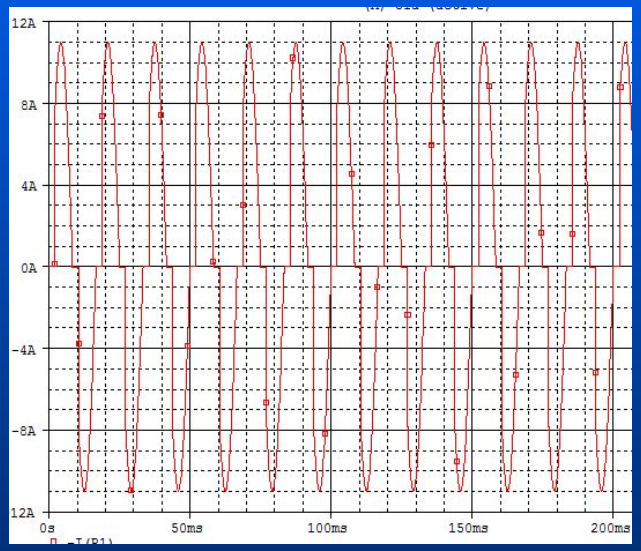
예제 5-1(2) Pspice Simulation

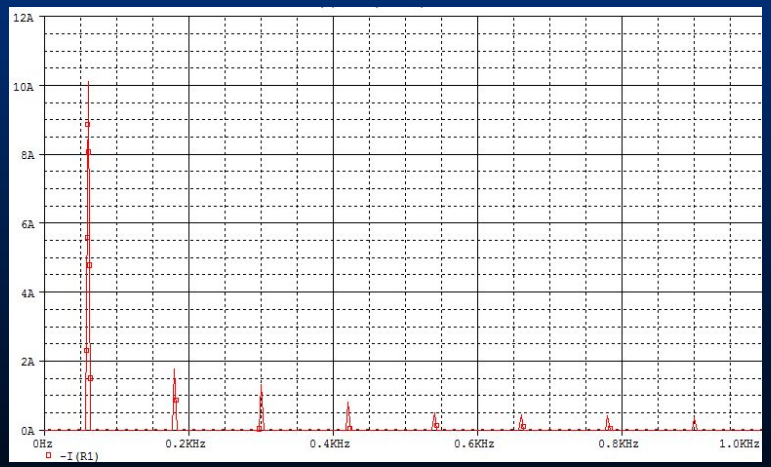


- *Parameter*
  - ✔ *R=15 Ω*
  - ✔ *f=60 Hz*
  - ✔ *α= 90*
    - *- TD1:0.0042(1/240)*
    - *- TD2:0.0125(3/240)*
  - ✔ *PW=0.001*
  - ✔ V_m*=100V*
  - ✔ *Transient Step*
    =0 50 ms
- *Find: i,* i_{avg}*, P*

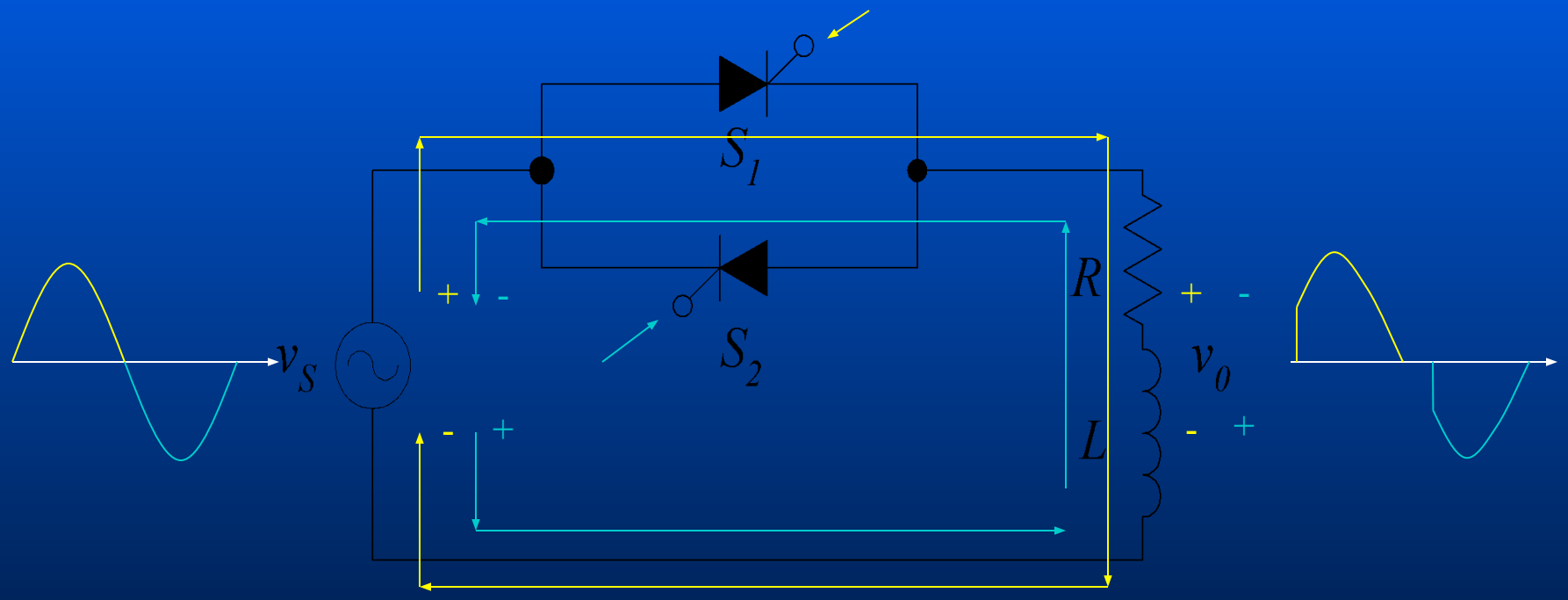
5.2 단상교류전압제어기: 고조파





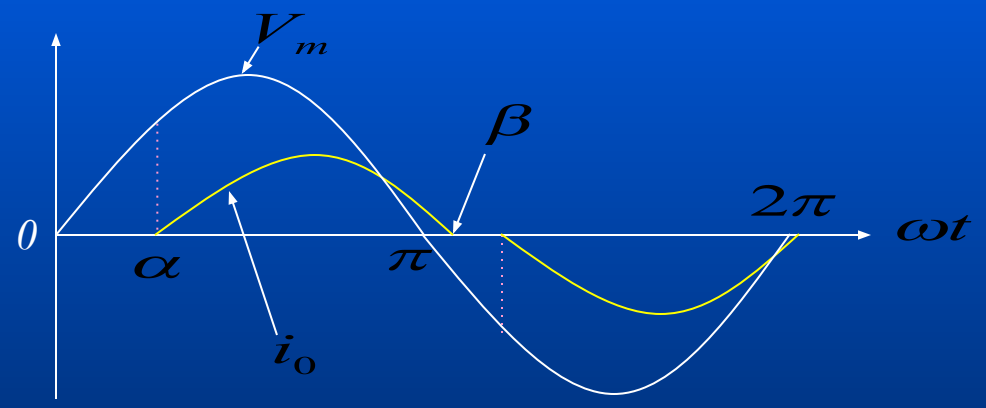


5.2 단상교류전압제어기: R-L 제어회로



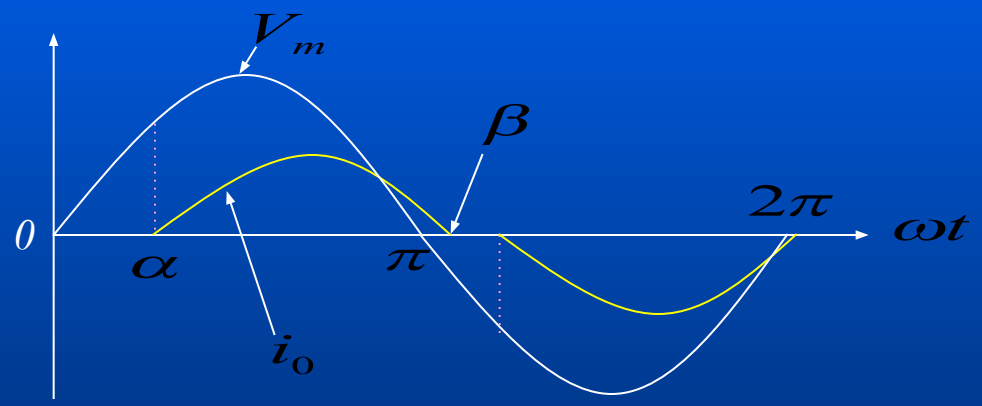
$$V_m \sin(\omega t) = R i_0(t) + L \frac{di_0(t)}{dt}$$

5.2 단상교류전압제어기: R-L부하 전류



$$i(\omega t) = \begin{cases} \dfrac{V_m}{Z}\left[\sin(\omega t - \theta) + \sin(\alpha - \theta)\cdot e^{-\frac{(\alpha - \omega t)R}{\omega L}}\right] & at\ \alpha \le \omega t \le \beta \\ 0 & otherwise \end{cases}$$

5.2 단상교류전압제어기: R-L부하 통전각



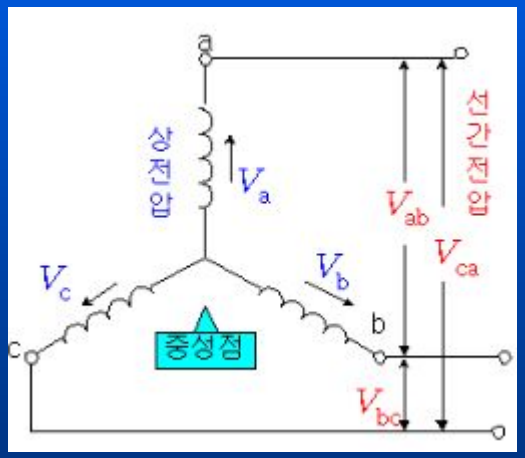
- 소호각(Extinction Angle)

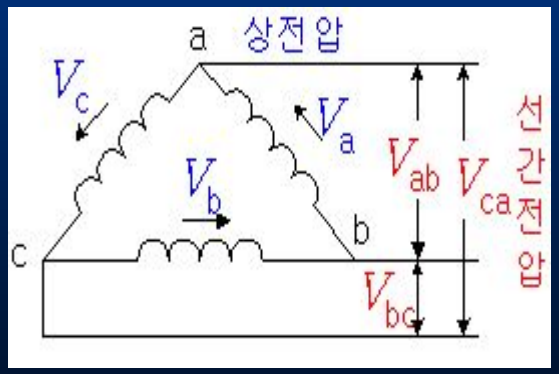
$$i(\beta) = 0 = \frac{V_m}{Z}\left[\sin(\beta-\theta) + \sin(\alpha-\theta)\cdot e^{-\frac{(\alpha-\beta)R}{\omega L}}\right]$$

- 통전각(Duration Angle)

$$\gamma = \beta - \alpha$$

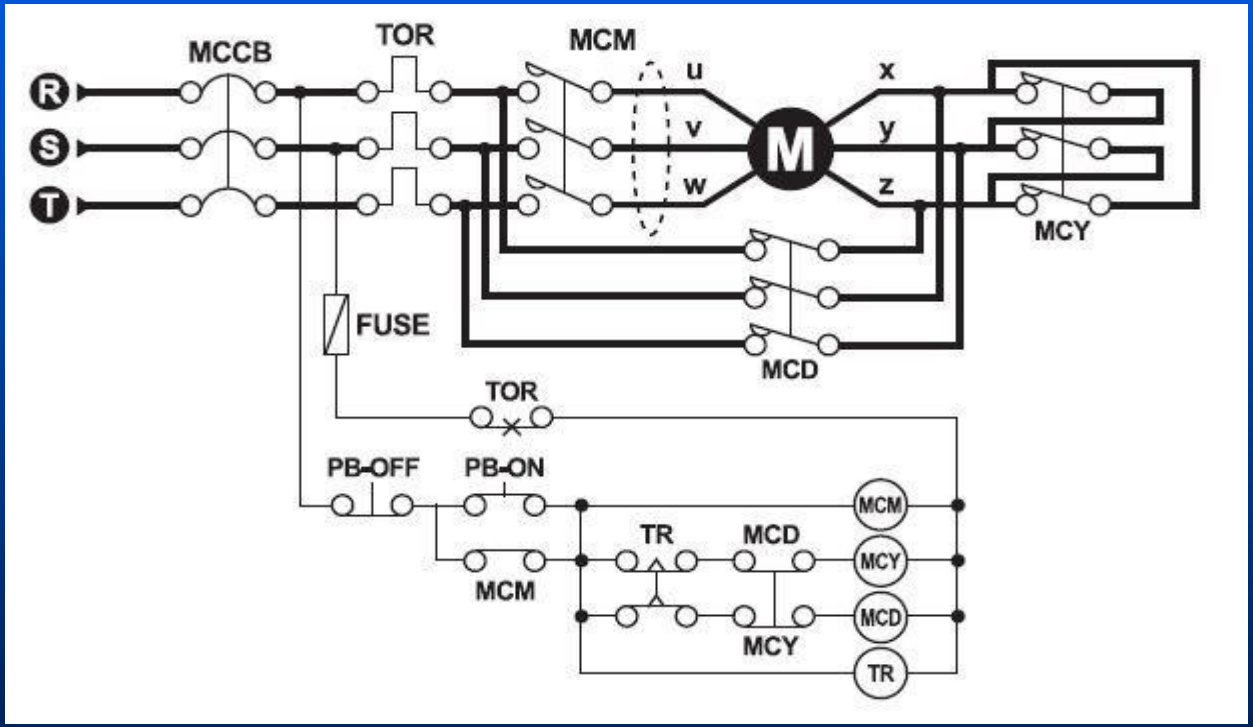
3상회로





결선	선간 전압	선간 전류	전력
Y	$\sqrt{3}$ × 상전압	상전류	$\sqrt{3}$ × 상전압 × 상전류
Δ	상전압	$\sqrt{3}$ × 상전류	$\sqrt{3}$ × 상전압 × 상전류

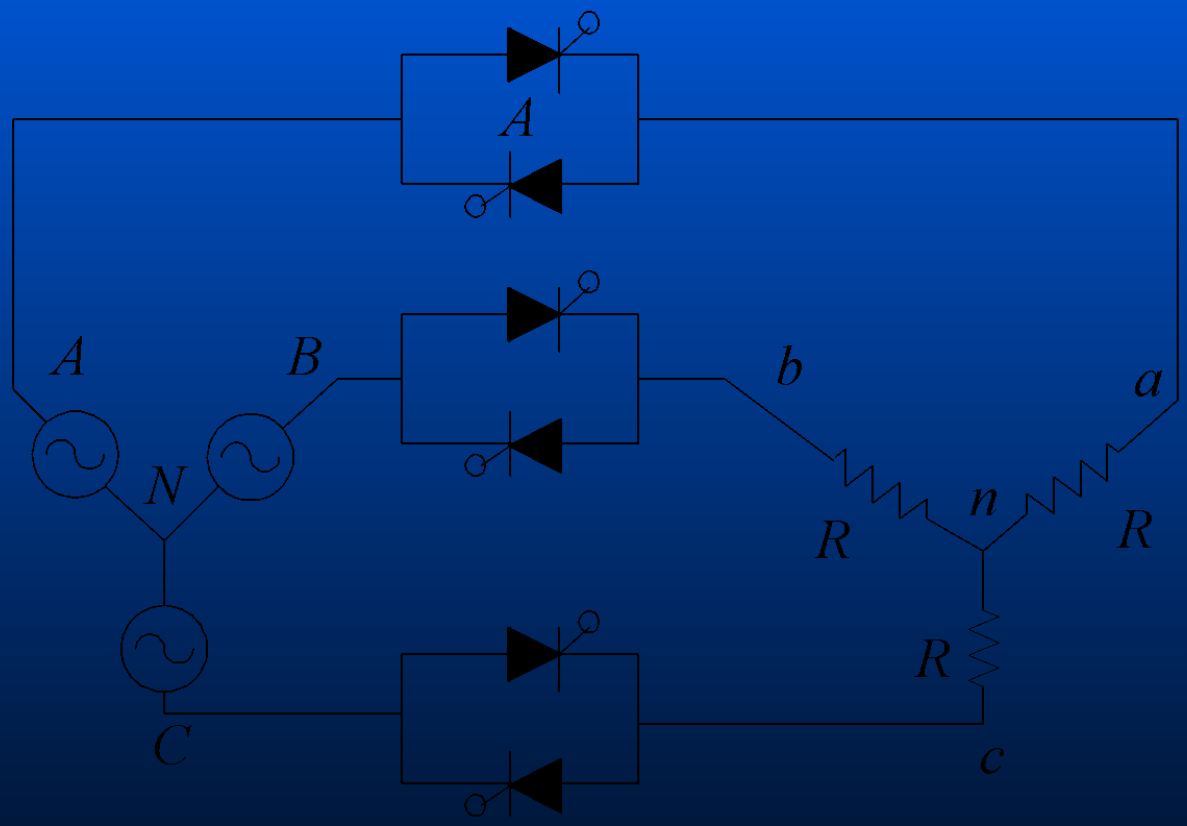
Y-Δ 기동법



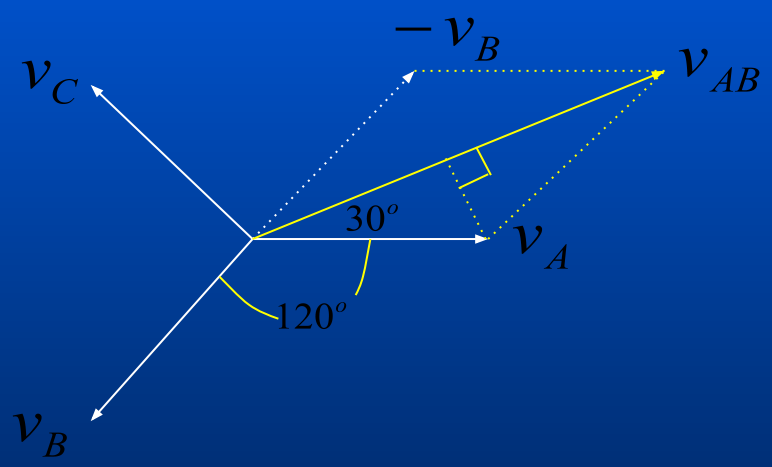
- 기동시: 전류를 줄이기 위해 Y 결선
- 운전시: 회전력을 높이기 위해 Δ결선

결선	선간전류	선전류	1상전력
Y	$\sqrt{3}$ × 상전압	상전류	$\sqrt{3}$ × 상전압 × 상전류
Δ	상전압	$\sqrt{3}$ × 상전류	$\sqrt{3}$ × 상전압 × 상전류

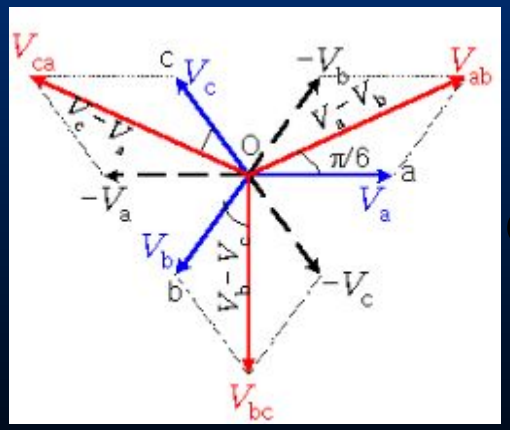
5.3 3상 전압조절기(Y 결선)

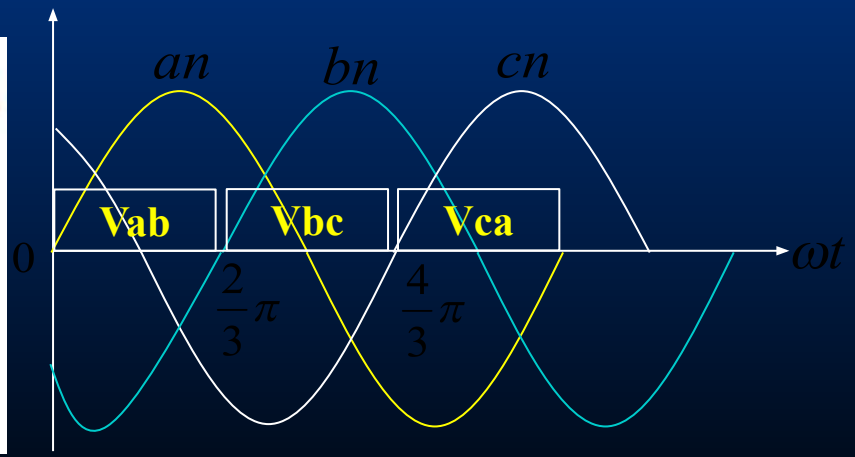


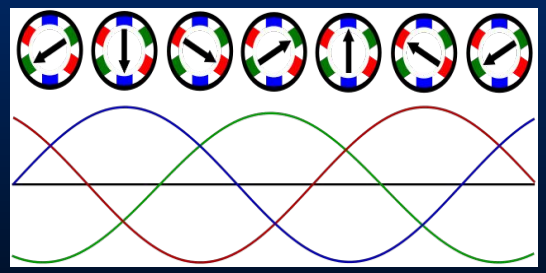
5.3 3상 회로: 전압



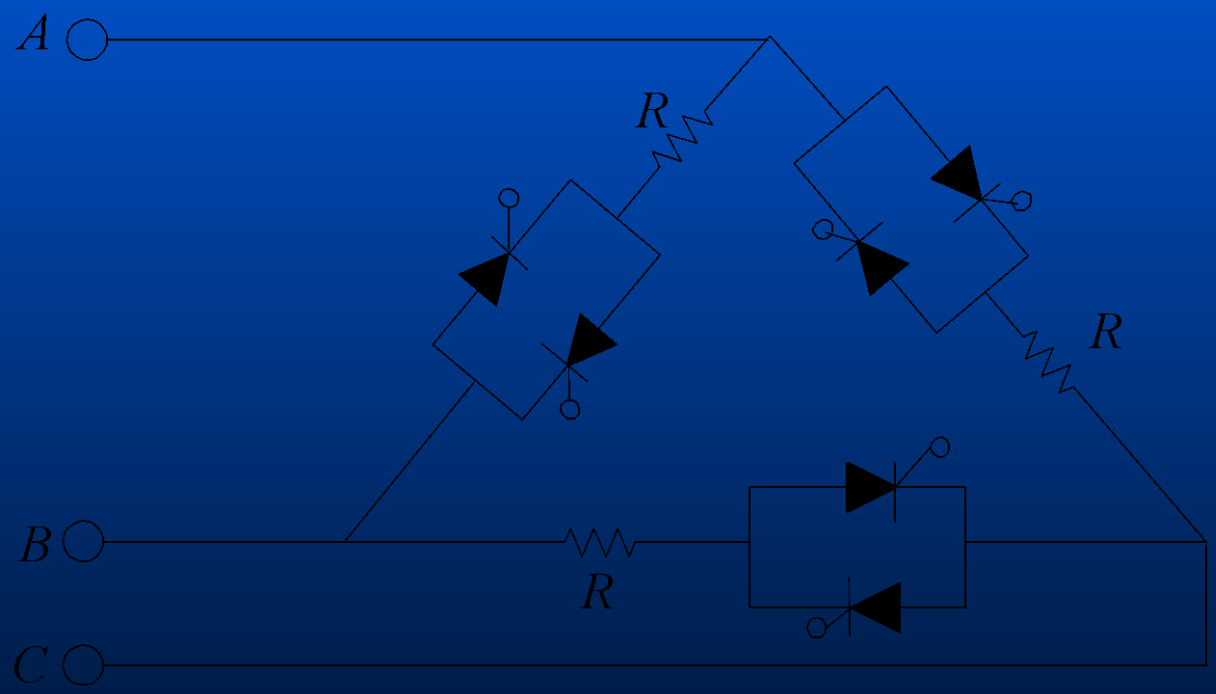
$$v_{AB} = 2v_A \cos(30^o)\angle 30^o$$
$$= 2v_A \frac{\sqrt{3}}{2}\angle 30^o$$
$$= \sqrt{3}v_A\angle 30^o$$







5.3 3상 전압조절기(Δ 결선)

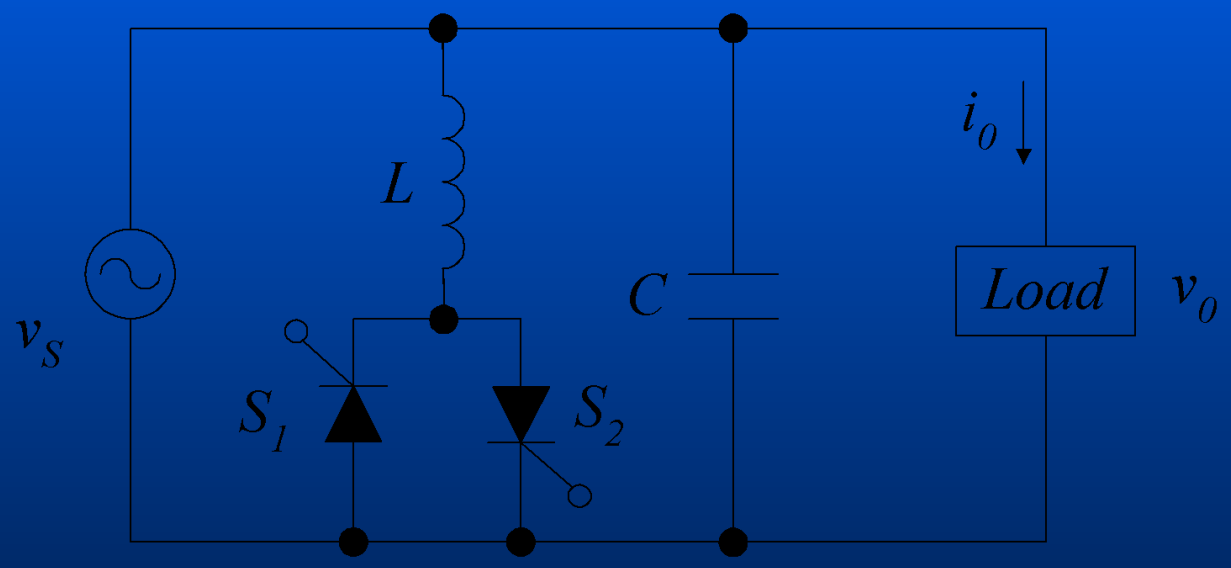


5.4 유도전동기 속도제어

- 전압제어(Voltage Control)
  - 1차 전압제어
  - Squirrel-cage(농형)
- 전압-주파수제어(Voltage /Frequency Control)
- 주파수 제어(Frequency Control)

$$N = \frac{120f}{P}(1-s)$$

5.5 정지형 무효전력제어



- L의 가변에 의한 역률 보상
- Load의 값에 따라 효율적으로 무효전력 관리